Abstract algebra homework3

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Problem 1

Let G be a group and $H \leq G$ a subgroup. Recall $Hx = \{hx : h \in H\}$.

(a)

Claim. *$H \trianglelefteq G$ if and only if $Hx = xH$ for all $x \in G$.*

Proof. By definition, H is normal in G means $xHx^{-1} = H$ for every $x \in G$.

($\Rightarrow$) Suppose $H \trianglelefteq G$, so $xHx^{-1} = H$ for all $x \in G$. Multiplying both sides on the right by x gives

$$xH = xHx^{-1}x = Hx.$$

($\Leftarrow$) Suppose $Hx = xH$ for all $x \in G$. Multiplying both sides on the right by x^{-1} gives

$$H = Hxx^{-1} = xHx^{-1}.$$

Since this holds for every $x \in G$, H is normal. □

(b)

Claim. *If $[G : H] = 2$, then $H \trianglelefteq G$.*

Proof. By part (a) it suffices to show $Hx = xH$ for every $x \in G$.

If $x \in H$, then $Hx = H$ (since H is closed under its own operation and H is a subgroup, right multiplication by $x \in H$ permutes H) and likewise $xH = H$, so $Hx = xH$.

If $x \notin H$, note that the right cosets of H partition G, and since $[G : H] = 2$ there are exactly two of them: H and Hx (as $x \notin H$, $Hx \neq H$). Since they partition G,

$$Hx = G \setminus H.$$

By the identical argument applied to left cosets, there are exactly two left cosets, H and xH, and

$$xH = G \setminus H.$$

Hence $Hx = G \setminus H = xH$.

In both cases $Hx = xH$, so by part (a), $H \trianglelefteq G$. □

Problem 2

(a)

Claim. *If $x, y \in G$ have coprime orders, then* $\mathrm{gp}(x) \cap \mathrm{gp}(y) = \{e\}$.

Proof. Let $m = |x|$ and $n = |y|$, so $\gcd(m, n) = 1$. Let $z \in \mathrm{gp}(x) \cap \mathrm{gp}(y)$. Since $\mathrm{gp}(x)$ is cyclic of order m, the order of z divides m (Lagrange's theorem applied to $\mathrm{gp}(x)$). Since $\mathrm{gp}(y)$ is cyclic of order n, the order of z also divides n. Hence $|z|$ divides $\gcd(m, n) = 1$, so $|z| = 1$, i.e., $z = e$. Therefore $\mathrm{gp}(x) \cap \mathrm{gp}(y) \subseteq \{e\}$, and since e is always in the intersection, $\mathrm{gp}(x) \cap \mathrm{gp}(y) = \{e\}$. □

(b)

Claim. *Suppose every subgroup of G is normal. If $x, y \in G$ have coprime orders, then $xy = yx$.*

Proof. Let $H = \mathrm{gp}(x)$ and $K = \mathrm{gp}(y)$; by hypothesis both are normal in G. Consider the commutator

$$c := x^{-1}y^{-1}xy \in G.$$

Since $K \trianglelefteq G$, we have $x^{-1}yx \in K$ (conjugates of elements of K stay in K). Writing

$$c = (x^{-1}y^{-1}x)y,$$

both $x^{-1}y^{-1}x = (x^{-1}yx)^{-1} \in K$ and $y \in K$, so $c \in K$ (as K is closed under multiplication).

Since $H \trianglelefteq G$, we have $y^{-1}xy \in H$. Writing

$$c = x^{-1}(y^{-1}xy),$$

both $x^{-1} \in H$ and $y^{-1}xy \in H$, so $c \in H$.

Thus $c \in H \cap K = \mathrm{gp}(x) \cap \mathrm{gp}(y)$, which equals $\{e\}$ by part (a). Hence $c = e$, i.e.,

$$x^{-1}y^{-1}xy = e \iff xy = yx.$$ □

Problem 3

Throughout, $N \trianglelefteq G$ and $\pi : G \to G/N$, $\pi(x) = xN$, is the quotient map.

(a)

Claim. *For $N \subseteq H \leq G$, the assignment $\Gamma(H) := H/N = \{hN : h \in H\} \subseteq G/N$ is a well-defined subgroup of G/N.*

Proof. First, $\Gamma(H)$ is a well-defined subset of G/N (it is simply the image $\pi(H)$). We check it is a subgroup:

- *Identity*: since $e \in H$, $N = eN \in \Gamma(H)$.
- *Closure*: if $h_1N, h_2N \in \Gamma(H)$ with $h_1, h_2 \in H$, then $(h_1N)(h_2N) = (h_1h_2)N \in \Gamma(H)$ since $h_1h_2 \in H$.
- *Inverses*: if $hN \in \Gamma(H)$ with $h \in H$, then $(hN)^{-1} = h^{-1}N \in \Gamma(H)$ since $h^{-1} \in H$.

Hence $\Gamma(H) = H/N$ is a subgroup of G/N, so Γ indeed lands in $\{K \leq G/N\}$. □

(b)

Claim. *For $K \leq G/N$, the assignment $\Delta(K) := \pi^{-1}(K)$ is a well-defined subgroup of G containing N.*

Proof. $\Delta(K) = \pi^{-1}(K)$ is, by definition, a well-defined subset of G (the preimage of K under the function π).

$N \subseteq \Delta(K)$: For $n \in N$, $\pi(n) = nN = N$, which is the identity of G/N and hence lies in K (as K is a subgroup). So $n \in \pi^{-1}(K) = \Delta(K)$.

$\Delta(K)$ *is a subgroup*: This is the general fact that preimages of subgroups under homomorphisms are subgroups.

- *Identity*: $\pi(e) = N$ is the identity of G/N, which lies in K, so $e \in \Delta(K)$.
- *Closure*: if $a, b \in \Delta(K)$, then $\pi(a), \pi(b) \in K$, so $\pi(ab) = \pi(a)\pi(b) \in K$ (as K is closed), hence $ab \in \Delta(K)$.
- *Inverses*: if $a \in \Delta(K)$, then $\pi(a^{-1}) = \pi(a)^{-1} \in K$ (as K is closed under inverses), so $a^{-1} \in \Delta(K)$.

Hence $\Delta(K) \leq G$ and $N \subseteq \Delta(K)$. □

(c)

Claim. *Γ and Δ are mutually inverse bijections between $\{H \leq G : N \subseteq H\}$ and $\{K \leq G/N\}$.*

Proof. We show $\Delta(\Gamma(H)) = H$ for all H with $N \subseteq H \leq G$, and $\Gamma(\Delta(K)) = K$ for all $K \leq G/N$.

$\Delta(\Gamma(H)) = H$**:** We must show $\pi^{-1}(H/N) = H$.

($\supseteq$) If $h \in H$, then $\pi(h) = hN \in H/N$, so $h \in \pi^{-1}(H/N)$.

($\subseteq$) If $x \in \pi^{-1}(H/N)$, then $\pi(x) = xN \in H/N$, so $xN = hN$ for some $h \in H$. Then $x = hn$ for some $n \in N$. Since $N \subseteq H$, we have $n \in H$, and hence $x = hn \in H$ (product of two elements of H).

Combining both inclusions, $\pi^{-1}(H/N) = H$, i.e., $\Delta(\Gamma(H)) = H$.

$\Gamma(\Delta(K)) = K$**:** We must show $\pi(\pi^{-1}(K)) = K$. This holds for the preimage/image of *any* subset K under a *surjective* map π:

($\subseteq$) is immediate: if $y \in \pi(\pi^{-1}(K))$, then $y = \pi(x)$ for some $x \in \pi^{-1}(K)$, so $y = \pi(x) \in K$.

($\supseteq$): let $k \in K \subseteq G/N$. Since π is surjective, there is $x \in G$ with $\pi(x) = k$. Since $k \in K$, we get $x \in \pi^{-1}(K)$, so $k = \pi(x) \in \pi(\pi^{-1}(K))$.

Hence $\pi(\pi^{-1}(K)) = K$, i.e., $\Gamma(\Delta(K)) = K$.

Since $\Gamma \circ \Delta = \mathrm{id}$ and $\Delta \circ \Gamma = \mathrm{id}$, Γ and Δ are mutually inverse bijections. □

(d)

Claim. *Let $N \subseteq H \leq G$. Then $H \trianglelefteq G$ if and only if $H/N \trianglelefteq G/N$, and in that case*

$$G/H \cong (G/N)/(H/N).$$

Proof. (⇒) Suppose $H \trianglelefteq G$. Let $xN \in G/N$ be arbitrary. Then

$$(xN)(H/N)(xN)^{-1} = \{(xN)(hN)(x^{-1}N) : h \in H\} = \{(xhx^{-1})N : h \in H\}.$$

As h ranges over H, xhx^{-1} ranges over $xHx^{-1} = H$ (since H is normal), so this set equals $\{h'N : h' \in H\} = H/N$. Since xN was arbitrary, $H/N \trianglelefteq G/N$.

(⇐) Suppose $H/N \trianglelefteq G/N$. Let $x \in G$ and $h \in H$; we show $xhx^{-1} \in H$. Applying π,

$$\pi(xhx^{-1}) = (xN)(hN)(xN)^{-1} \in (xN)(H/N)(xN)^{-1} = H/N,$$

using normality of H/N in G/N. So $\pi(xhx^{-1}) = h'N$ for some $h' \in H$, i.e., $xhx^{-1}N = h'N$, so $xhx^{-1} = h'n$ for some $n \in N$. Since $N \subseteq H$, $n \in H$, so $xhx^{-1} = h'n \in H$. As x, h were arbitrary, $H \trianglelefteq G$.

Isomorphism. Assume $H \trianglelefteq G$ (equivalently $H/N \trianglelefteq G/N$). Let $\rho : G/N \to (G/N)/(H/N)$ be the canonical quotient map, and define

$$\psi := \rho \circ \pi : G \longrightarrow (G/N)/(H/N), \qquad \psi(x) = \pi(x)(H/N) = (xN)(H/N).$$

As a composite of two group homomorphisms, ψ is a homomorphism, and it is surjective since both π and ρ are surjective.

We compute $\ker\psi$: $\psi(x) =$ identity of $(G/N)/(H/N)$ iff $\pi(x) = xN \in H/N$ iff $xN = hN$ for some $h \in H$ iff $x = hn$ for some $h \in H, n \in N \subseteq H$ iff $x \in H$. So $\ker\psi = H$.

By the First Isomorphism Theorem applied to ψ,

$$G/H = G/\ker\psi \;\cong\; \mathrm{im}\,\psi = (G/N)/(H/N).$$ □

Problem 4

(a)

Claim. *The map $\alpha : G \times G/H \to G/H$, $\alpha(x, yH) = (xy)H$, is well-defined and is a group action of G on G/H.*

Proof. **Well-defined.** Suppose $yH = y'H$, i.e., $y' = yh$ for some $h \in H$. Then

$$(xy')H = (xyh)H = (xy)(hH) = (xy)H,$$

since $hH = H$ for $h \in H$. So $\alpha(x, yH)$ does not depend on the choice of representative y, and α is well-defined.

Action axioms.

- *Identity*: $\alpha(e, yH) = (ey)H = yH$ for all $yH \in G/H$.
- *Compatibility*: for $x_1, x_2 \in G$,

$$\alpha(x_1, \alpha(x_2, yH)) = \alpha(x_1, (x_2y)H) = (x_1(x_2y))H = ((x_1x_2)y)H = \alpha(x_1x_2, yH),$$

  using associativity in G.

Hence α defines a (left) group action of G on G/H. □

(b)

Claim. *If H is a subgroup of finite index in G, there exists a normal subgroup $N \trianglelefteq G$ with $N \subseteq H$, of finite index in G; in particular, if H is a proper subgroup of G, so is N.*

Proof. Let $n = [G : H] < \infty$ and let $X = G/H$, the set of left cosets, $|X| = n$. By part (a), the action α of G on X by $\alpha(x, yH) = (xy)H$ gives, for each $x \in G$, a permutation $\alpha_x : X \to X$, $\alpha_x(yH) = (xy)H$, and the map

$$\varphi : G \to \mathrm{Sym}(X), \qquad \varphi(x) = \alpha_x$$

is a group homomorphism (this is the standard permutation representation associated to a group action; $\varphi(x_1x_2) = \alpha_{x_1x_2} = \alpha_{x_1} \circ \alpha_{x_2} = \varphi(x_1)\varphi(x_2)$ by the compatibility axiom shown in (a)). Fixing a bijection $X \leftrightarrow \{1, \dots, n\}$, we may regard $\mathrm{Sym}(X) \cong S_n$.

Let

$$N := \ker \varphi = \{x \in G : xyH = yH \text{ for all } y \in G\} = \bigcap_{y \in G} yHy^{-1},$$

the *core* of H in G. As the kernel of a homomorphism, $N \trianglelefteq G$.

$N \subseteq H$: taking $y = e$ in the intersection, $N \subseteq eHe^{-1} = H$.

Finite index: by the First Isomorphism Theorem, $G/N \cong \operatorname{im} \varphi \leq \mathrm{Sym}(X) \cong S_n$, so

$$[G : N] = |G/N| = |\operatorname{im} \varphi| \leq n! < \infty.$$

Finally, since $N \subseteq H$, if H is a proper subgroup of G (i.e. $H \neq G$), then $N \subseteq H \neq G$ forces $N \neq G$ as well, so N is a proper (nontrivial in the sense of $N \neq G$) normal subgroup of finite index. □

Problem 5

We compute disjoint cycle decompositions in S_7, composing permutations right-to-left (i.e., apply the rightmost factor first).

(a)

$$(17)(16)(15)(14)(13)(12)$$

Tracking each element (applying (12) first, then (13), ..., then (17) last):

$$1 \mapsto 2, \quad 2 \mapsto 3, \quad 3 \mapsto 4, \quad 4 \mapsto 5, \quad 5 \mapsto 6, \quad 6 \mapsto 7, \quad 7 \mapsto 1.$$

$$\boxed{(17)(16)(15)(14)(13)(12) = (1234567)}$$

(b)

$$(12)(245)(12)$$

This is a conjugate of the cycle (245) by $\tau = (12)$ (note $\tau^{-1} = \tau$). Using the conjugation formula $\tau(a_1\, a_2\, \cdots\, a_k)\tau^{-1} = (\tau(a_1)\, \tau(a_2)\, \cdots\, \tau(a_k))$:

$$(12)(245)(12)^{-1} = (\tau(2)\, \tau(4)\, \tau(5)) = (1\,4\,5).$$

$$\boxed{(12)(245)(12) = (145)}$$

(c)

$$(123)(234)(456)$$

Tracking each element (applying (456) first, then (234), then (123) last):

$$1 \mapsto 2,\ 2 \mapsto 1,\ 3 \mapsto 4,\ 4 \mapsto 5,\ 5 \mapsto 6,\ 6 \mapsto 3,\ 7 \mapsto 7.$$

$$\boxed{(123)(234)(456) = (12)(3456)}$$

(d)

$$(1753)(27)(246)(1357)$$

Tracking each element (applying (1357) first, then (246), then (27), then (1753) last):

$$1 \mapsto 1,\ 2 \mapsto 4,\ 3 \mapsto 3,\ 4 \mapsto 6,\ 5 \mapsto 2,\ 6 \mapsto 5,\ 7 \mapsto 7.$$

$$\boxed{(1753)(27)(246)(1357) = (2465)}$$

(fixing $1, 3, 7$).

(e)

$$(123)(234)(45)(567)(74)$$

Tracking each element (applying (74) first, then (567), then (45), then (234), then (123) last):

$$1 \mapsto 2,\ 2 \mapsto 1,\ 3 \mapsto 4,\ 4 \mapsto 3,\ 5 \mapsto 6,\ 6 \mapsto 7,\ 7 \mapsto 5.$$

$$\boxed{(123)(234)(45)(567)(74) = (12)(34)(567)}$$

Problem 6

For a permutation in S_n with cycle type having n_j cycles of length j (for $j = 1, \ldots, n$, including fixed points as 1-cycles, so $\sum_j j\, n_j = n$), the centralizer has order $\prod_j j^{n_j}\, n_j!$, so the conjugacy class has size

$$\frac{n!}{\prod_j j^{n_j}\, n_j!}.$$

Here $n = 7$, so $n! = 5040$. We record the cycle type (in S_7, including fixed points) of each permutation from Problem 5 and compute its class size.

Permutation	Cycle type in S_7	Centralizer order	Class size
(a) (1234567)	$[7]$	$7^1 \cdot 1! = 7$	$5040/7 = 720$
(b) (145)	$[3,1,1,1,1]$	$3 \cdot 1! \cdot 4! = 72$	$5040/72 = 70$
(c) (12)(3456)	$[4,2,1]$	$4 \cdot 2 \cdot 1 = 8$	$5040/8 = 630$
(d) (2465)	$[4,1,1,1]$	$4 \cdot 3! = 24$	$5040/24 = 210$
(e) (12)(34)(567)	$[3,2,2]$	$3 \cdot 2^2 \cdot 2! = 24$	$5040/24 = 210$

Pairwise non-conjugacy. Two permutations in S_n are conjugate if and only if they have the same cycle type. The five cycle types above,

$$[7], \quad [3,1,1,1,1], \quad [4,2,1], \quad [4,1,1,1], \quad [3,2,2],$$

are pairwise distinct as partitions of 7 (note (d) and (e) have the same class size, 210, but different cycle types, hence different, non-conjugate classes). Therefore the five given permutations lie in five distinct conjugacy classes of S_7 and are pairwise not conjugate.

Problem 7

(a)

Claim. $A_n \trianglelefteq S_n$.

Proof. The sign map $\operatorname{sgn} : S_n \to \{\pm 1\}$ is a group homomorphism (where $\{\pm 1\}$ is the multiplicative group of order 2), since for $\sigma, \tau \in S_n$,

$$\operatorname{sgn}(\sigma\tau) = \operatorname{sgn}(\sigma)\operatorname{sgn}(\tau)$$

(the sign of a product of transpositions is the product of the signs, as sign counts the parity of the number of transpositions used, and these numbers add). By definition, $A_n = \ker(\operatorname{sgn})$. The kernel of any group homomorphism is a normal subgroup of the domain, so $A_n \trianglelefteq S_n$. □

(b)

We must count the conjugacy classes of S_6 that consist entirely of even permutations (i.e., that are contained in A_6).

Conjugacy classes of S_6 correspond to partitions of 6; there are $p(6) = 11$ of them:

$$[6],\ [5,1],\ [4,2],\ [4,1,1],\ [3,3],\ [3,2,1],\ [3,1,1,1],\ [2,2,2],\ [2,2,1,1],\ [2,1,1,1,1],\ [1^6].$$

A permutation with cycle type having k cycles (counting fixed points) is a product of $6-k$ transpositions, hence *even* iff $6-k$ is even, i.e. iff k is even. We tabulate $k =$ number of parts for each partition:

Cycle type	# parts k	even? (k even)
$[6]$	1	no
$[5,1]$	2	yes
$[4,2]$	2	yes
$[4,1,1]$	3	no
$[3,3]$	2	yes
$[3,2,1]$	3	no
$[3,1,1,1]$	4	yes
$[2,2,2]$	3	no
$[2,2,1,1]$	4	yes
$[2,1,1,1,1]$	5	no
$[1^6]$	6	yes

Exactly the 6 cycle types $[5,1], [4,2], [3,3], [3,1,1,1], [2,2,1,1], [1^6]$ consist of even permutations, i.e., are contained in A_6.

There are 6 conjugacy classes of S_6 contained in A_6.